

Team MorAura

Automated Soil Condition Monitoring System



Agenda

Project Overview

Part 1

What is MorAura

04

System Architecture

05

Hardware Design

06

Phase 1

Part 2

Demonstration

08

Bottlenecks

09

Phase 2

Part 3

New System Architecture

11

Data flow Design

12

Project Overview

Problem Statement

Monitoring plants is a tedious labour intensive task, and strict monitoring can reduce the likelihood of plants dying out and improve the yield of crops.

Our system is designed to monitor plant conditions to make sure they're within the ideal ranges.



System Architecture

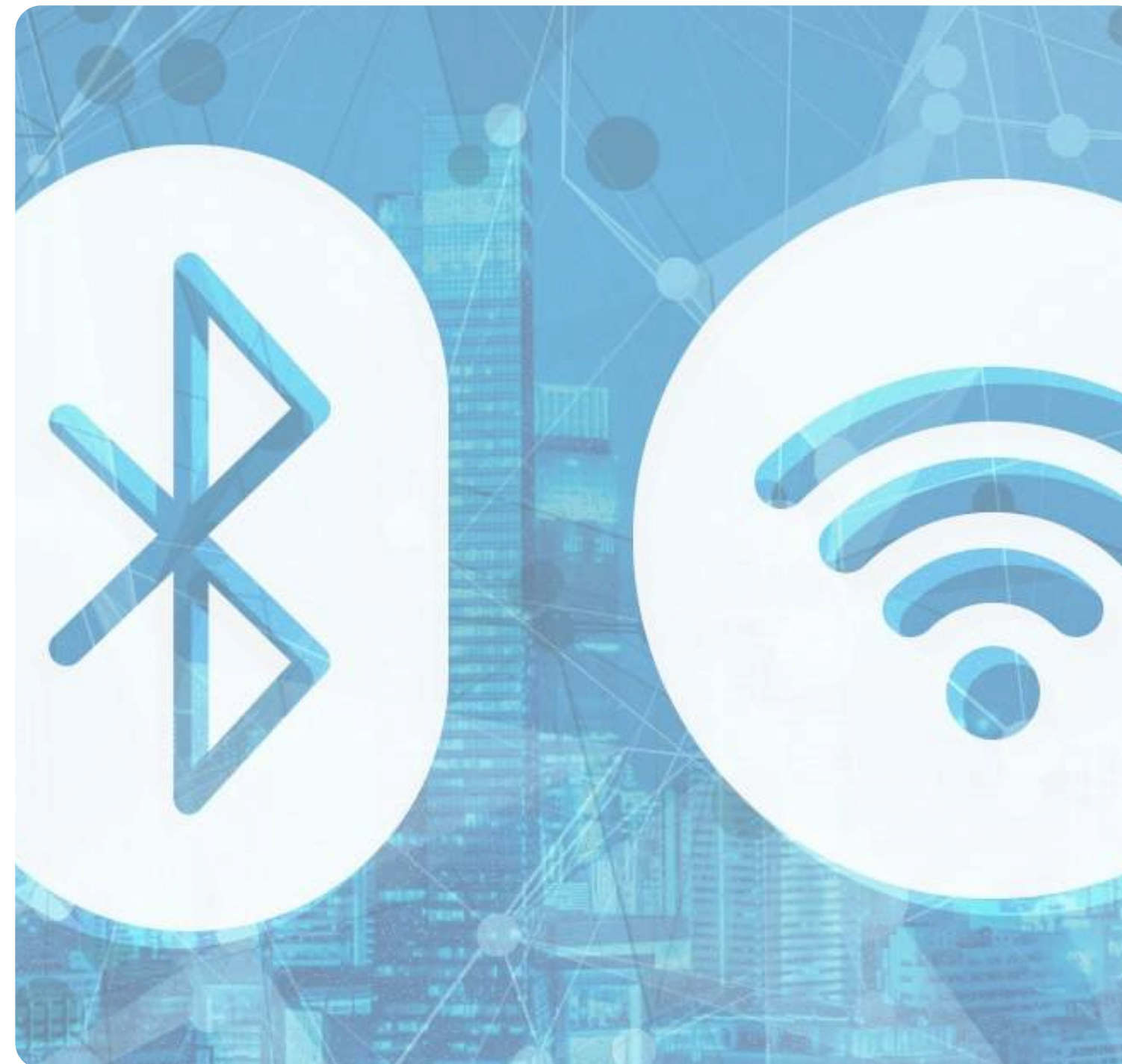
Data Collection

Log Light Moisture and Temperature



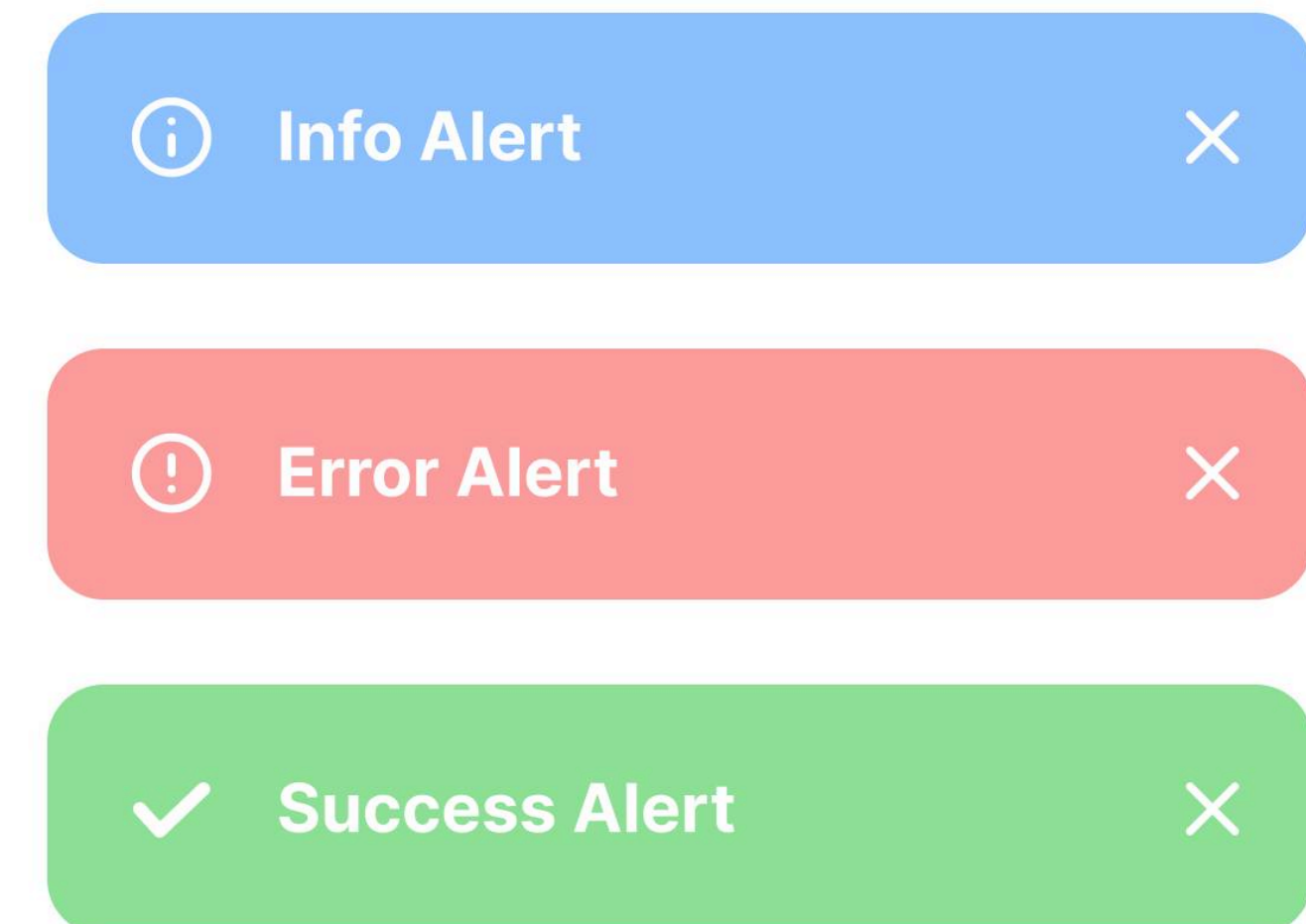
Communication

BLE / WiFi



User Experience

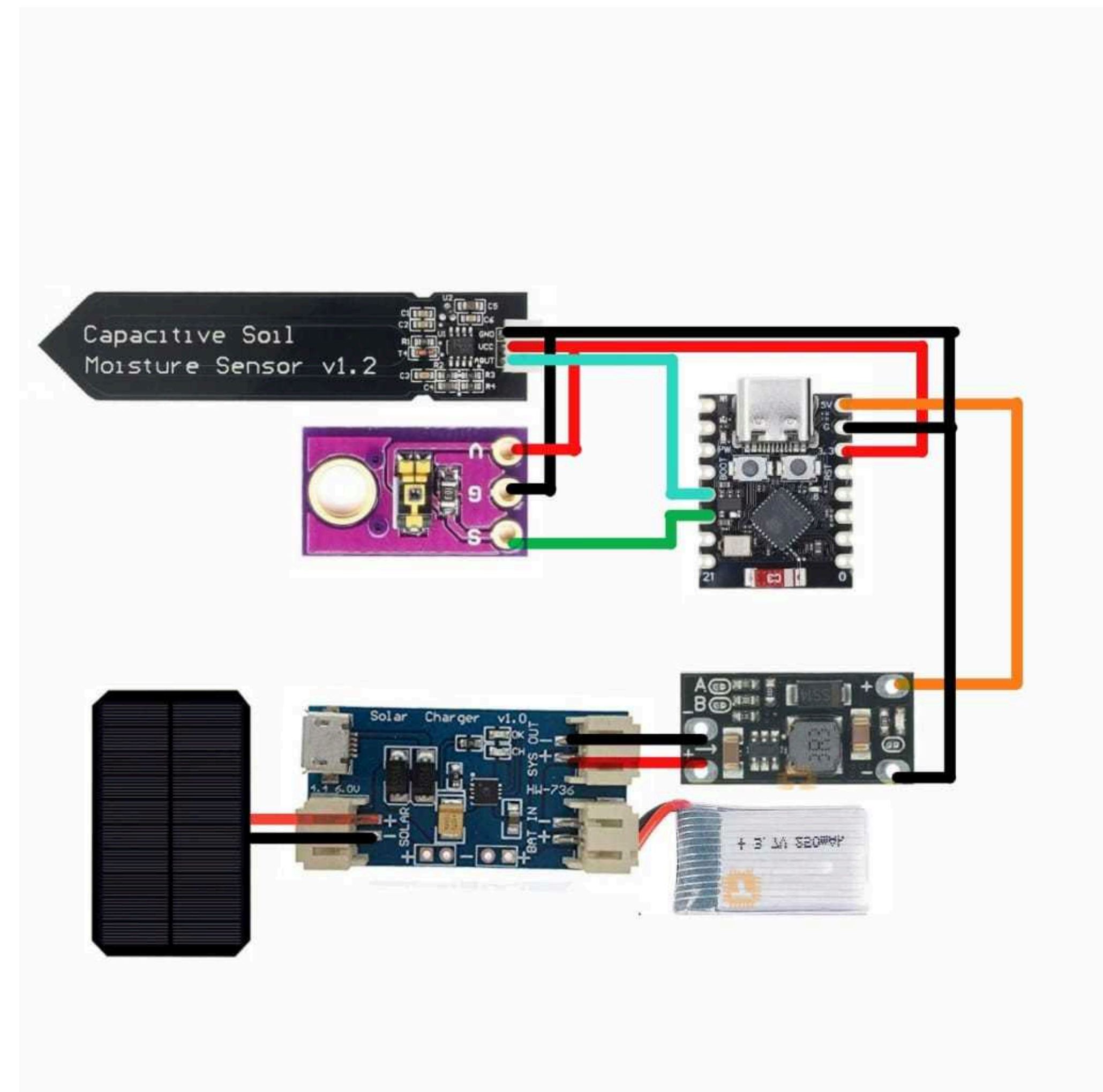
Show data and alerts



Hardware Design

ESP32 controller

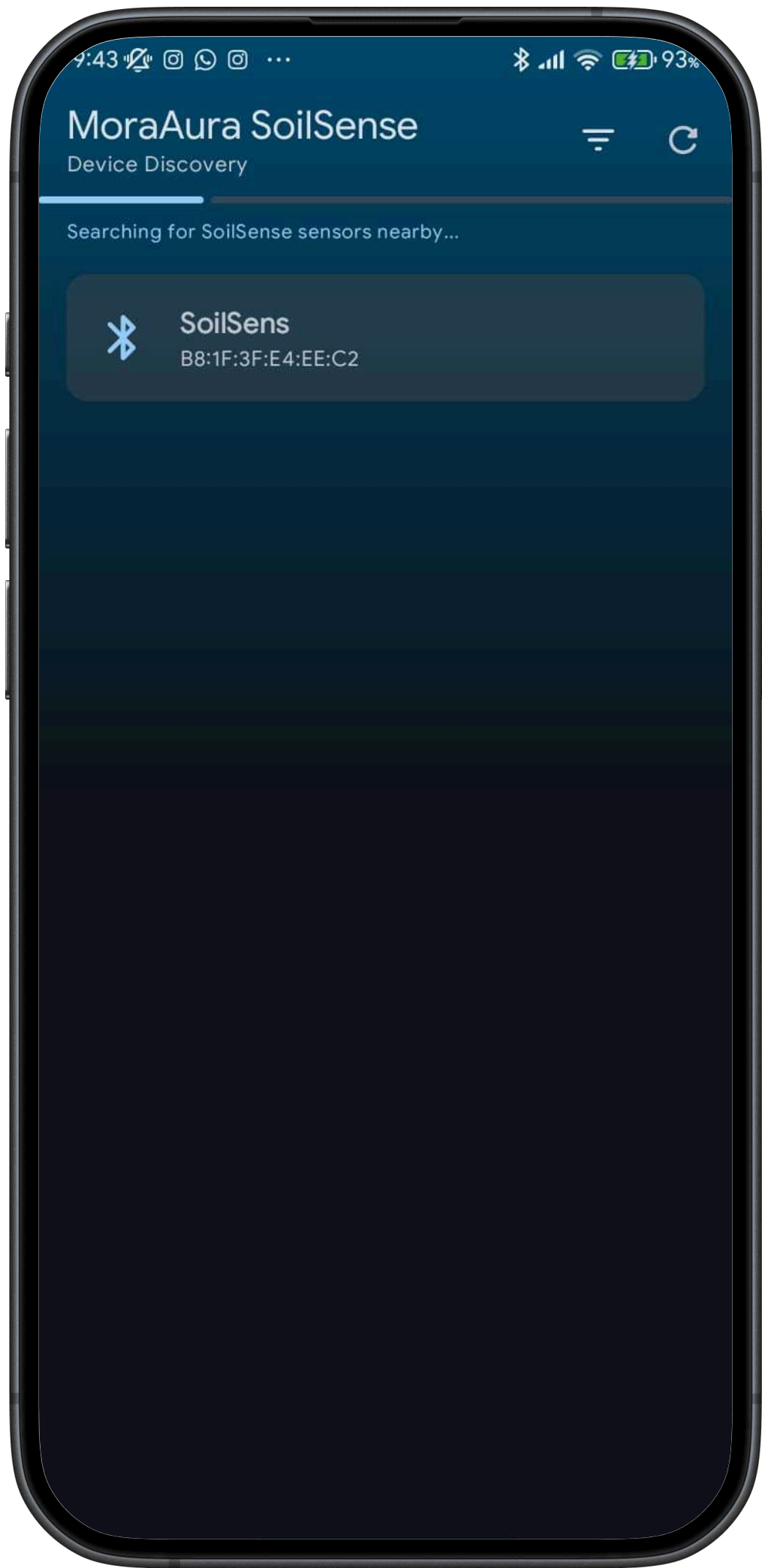
- Power system:
 - Solar charging module
 - 5 V mini solar panel
 - LiPo boost board
 - 3.7 V 250 mAh LiPo battery
- Sensors:
 - Capacitive soil moisture sensor
 - Ambient light sensor
 - Temperature is currently read from the ESP32'



Phase I

App Demonstration

Concept A



Feature 1
Describe feature

Feature 2
Describe feature

Feature 3
Describe feature

Bottlenecks

As a result of using BLE for communication the mobile device needs to be within range in order to send and receive data.

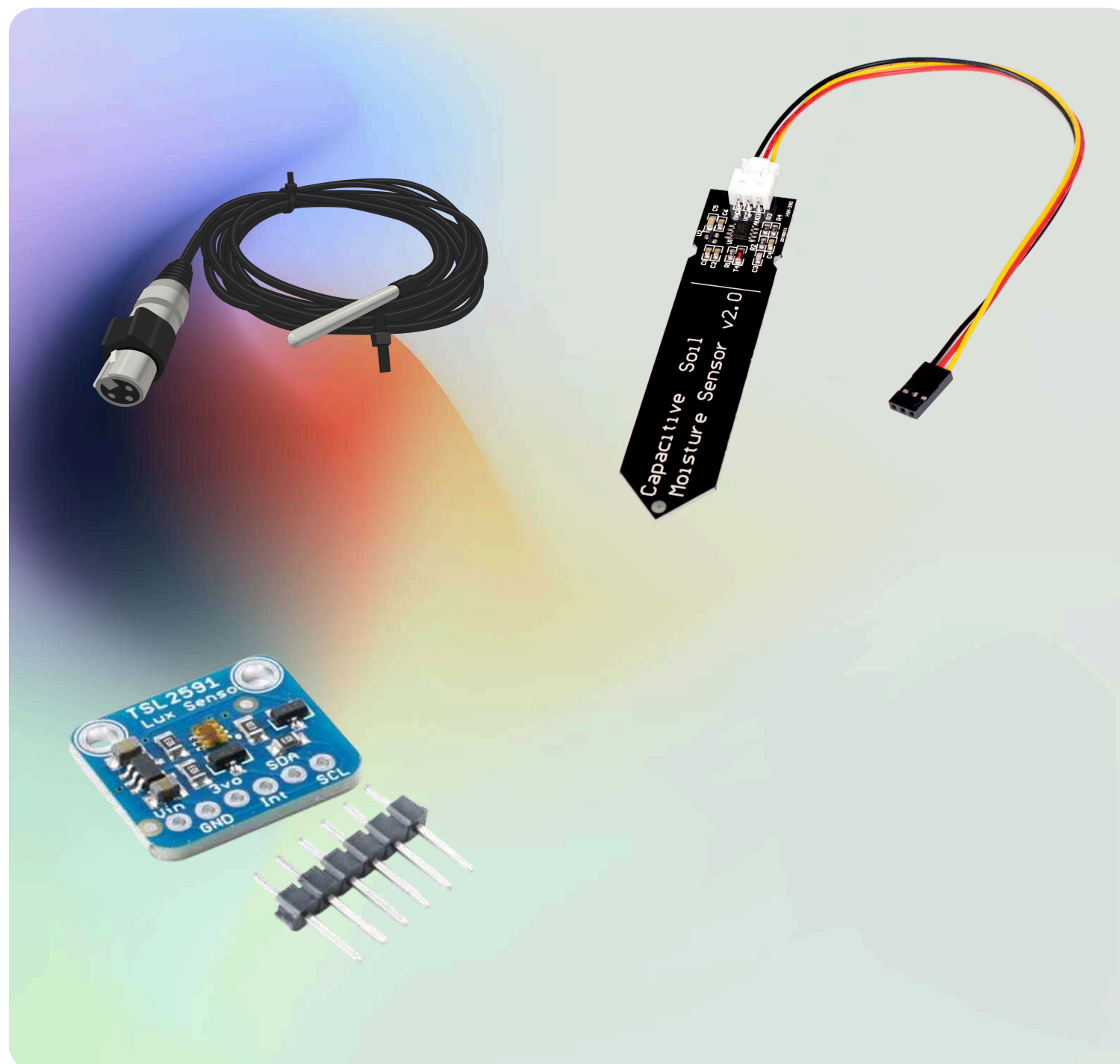
This hinders the ability to send out real time alerts on critical conditions.

Phase II

Improved System Architecture

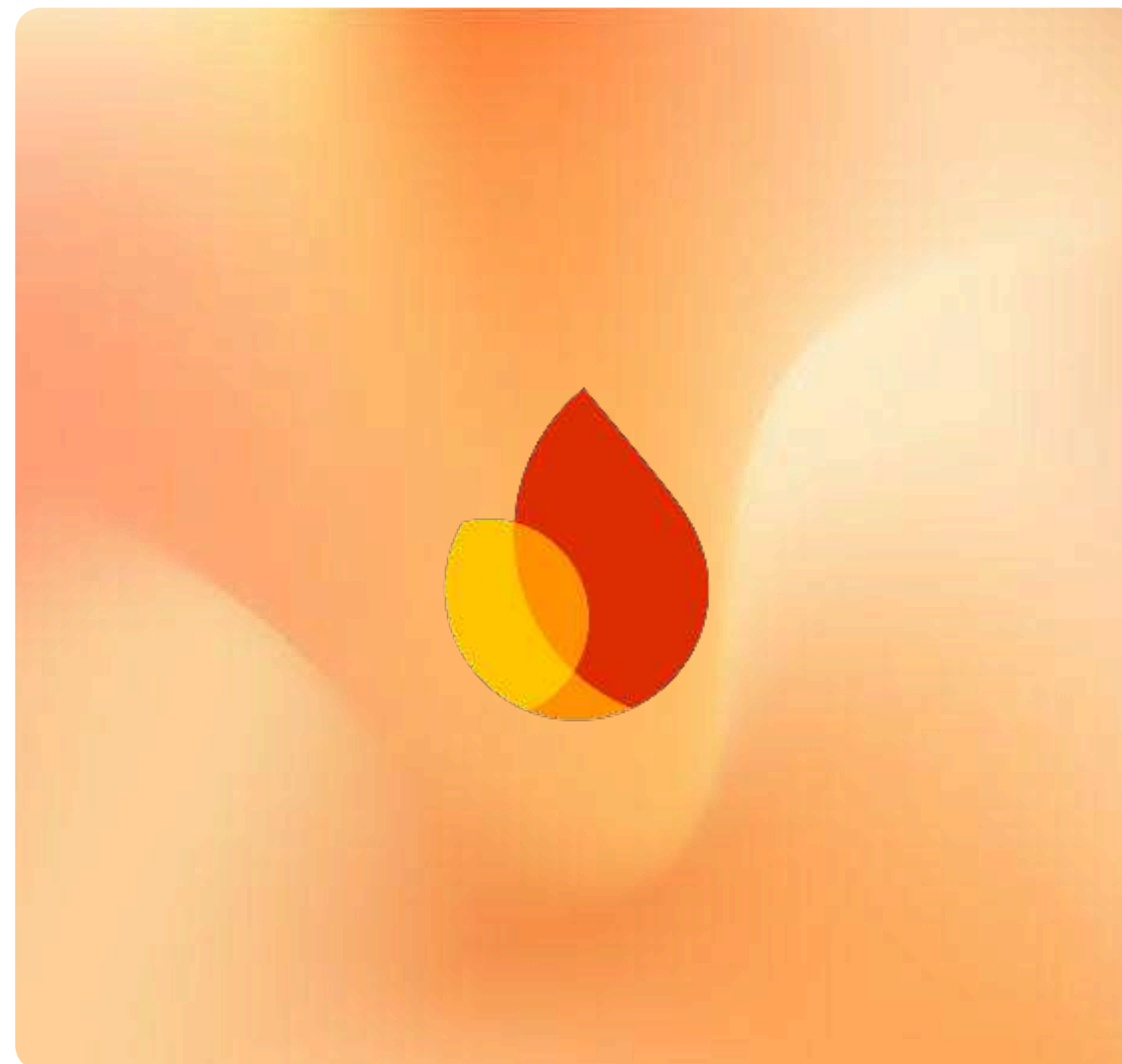
Data Collection

Same as in Phase 1



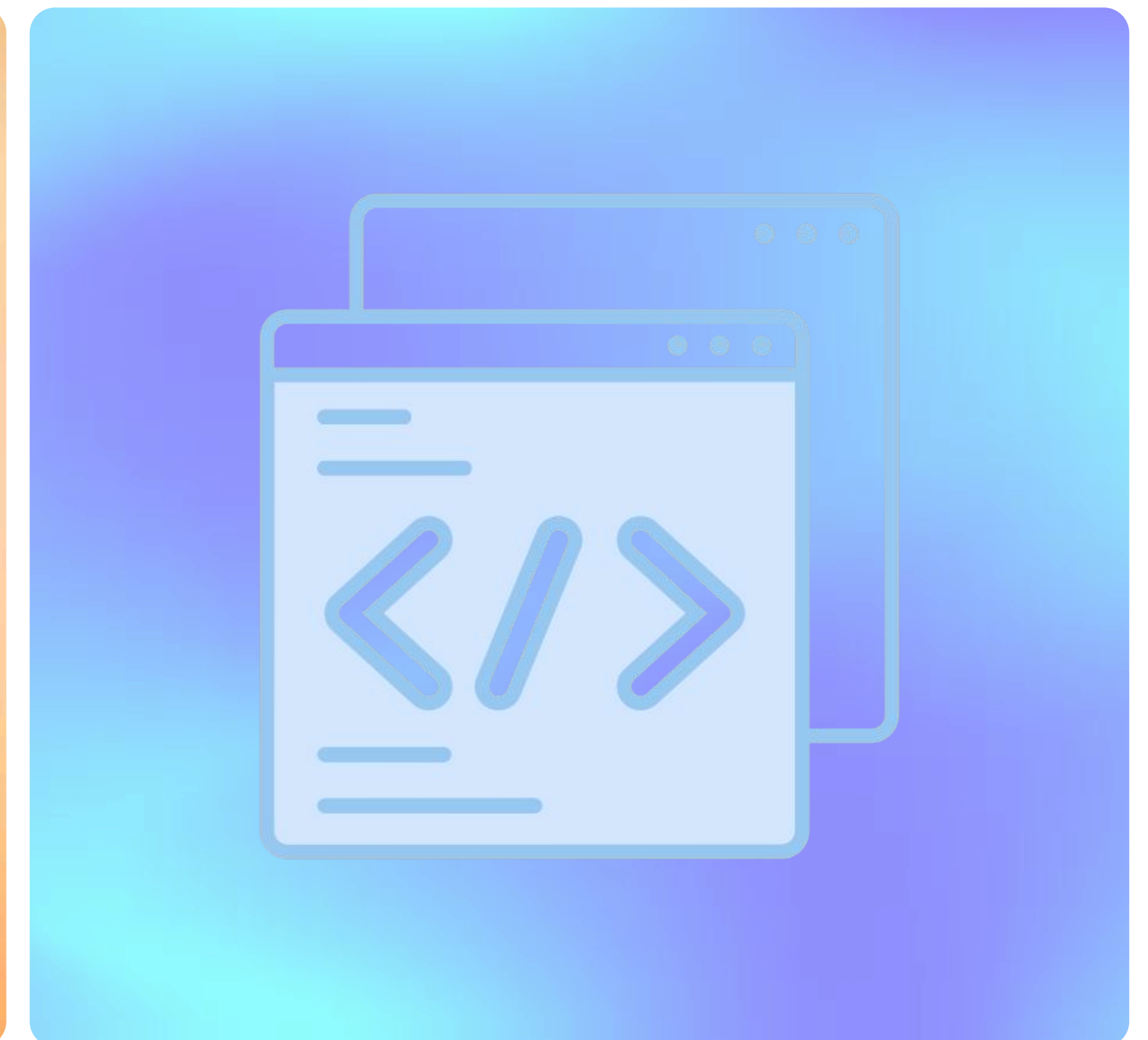
Database

Connect and upload data



User Experience

Access data via WebApp



Data flow design

The architecture begins with a separate configuration flow where the ESP32 acts as a local web server (captive portal) to securely capture initial Wi-Fi settings from the user. Once online, every time the ESP32 connects to the network, it syncs its internal time and uses standard REST API calls to fetch operational thresholds directly from the Firebase Real-time Database. Users manage these thresholds via a mobile app that pushes updates to the database for the hardware to retrieve on its next cycle. When specific conditions are met, database changes trigger Firebase Cloud Functions to generate alerts, which are then fetched and displayed as real-time notifications by a connected web application.

Thank you!

Team Members

Sudhira Jayatillaka
Minuka Gunawansa
Sathmi Bamunisinghe
Nadin Dharmawardena

Bhashitha Weerasingha
Sithuli Dinagee
Wenura Tharushika